Estimating Population Size Using Order Statistics

Mark Levene
School of Computing and Mathemtical Sciences,
Birkbeck, University of London
Malet Street, London WC1E 7HX, U.K.
m.levene@bbk.ac.uk

October 22, 2025

1 Introduction

Estimating the total size n of a population from a sample is a classical problem in statistics. When the full population is too large to observe, we rely on *order-statistics-based methods*. Two principal approaches are implemented:

- **Rank Inversion Estimator**
- **Spacing Estimator**

These methods leverage the information contained in the relative positions of sampled values (order statistics) to infer the total population size. Additionally, the concept of *pooling* multiple samples can improve the accuracy and reduce variability.

2 Problem Setup

Let the population be

$$\{X_1, X_2, \ldots, X_n\},$$

of unknown size n. We draw a sample of size $m \ll n$ without replacement. The goal is to estimate n using only the sample and an assumption about the underlying population distribution.

Two population models are considered:

1. **Normal distribution:** $X_i \sim \mathcal{N}(\mu, \sigma^2)$
2. **Pareto distribution:** $X_i \sim \text{Pareto}(\alpha, x_m)$

The MATLAB script draws repeated samples (n_{reps} times) to evaluate the variability and bias of the estimators.

3 Order Statistics

Let

$$X_{(1)} \leq X_{(2)} \leq \cdots \leq X_{(m)}$$

denote the sorted sample values (order statistics). Key properties:

1. **Distribution of order statistics:** For a continuous cumulative distribution function (CDF) $F(x)$,

$$P(X_{(i)} \leq x) = \sum_{k=i}^{m} \binom{m}{k} [F(x)]^k [1 - F(x)]^{m-k}.$$

2. **Relationship to population size:** In large populations, the rank of a value in the sample approximately represents its fraction of the total population. This principle underlies the rank inversion method.

4 Rank Inversion Estimator

4.1 Concept

The rank inversion estimator uses the *empirical rank* of an ordered sample value and inverts the *CDF* to estimate the population size.

4.2 Method

For the i-th order statistic $X_{(i)}$, the fraction of the population below it is approximated by the CDF:

$$F(X_{(i)}) \approx \frac{i}{n}.$$

Rearranging gives an estimate of n:

$$\widehat{n}_i = \frac{i - 0.5}{F(X_{(i)})},$$

where the subtraction of 0.5 is a *continuity correction* to reduce bias in small samples.

To reduce sensitivity to extreme values, the MATLAB script considers only the *central 25%–75% of order statistics*:

$$i_{\text{idx}} = \text{floor}(0.25m) : \text{ceil}(0.75m).$$

The final estimate is taken as the *median* over these indices:

$$\widehat{n}_{\text{rank}} = \text{median}\{\widehat{n}_i\}_{i \in i_{\text{idx}}}.$$

4.3 Justification

- The CDF maps sample ranks to fractions of the population.
- Using the median provides robustness against outliers and skewed data.

5 Spacing Estimator

5.1 Concept

The spacing estimator is based on the difference between consecutive order statistics and the local density of the distribution. Intuitively, in regions of high density, spacings are small, while in low-density regions, spacings are larger. The estimator leverages this relationship.

5.2 Method

For consecutive order statistics $X_{(i)}, X_{(i+1)}$, define the spacing:

$$\Delta_i = X_{(i+1)} - X_{(i)}.$$

If the underlying probability density function (PDF) is $f(x)$, the expected spacing is approximately:

$$E[\Delta_i] \approx \frac{1}{nf\left(\frac{X_{(i)}+X_{(i+1)}}{2}\right)}.$$

Rearranging gives an estimate of n:

$$\widehat{n}_i = \frac{1}{\Delta_i\, f\left(\frac{X_{(i)}+X_{(i+1)}}{2}\right)}.$$

Again, only the *central 25%–75% of spacings* are used, and the median of $\widehat{n}_i$ provides a robust estimate:

$$\widehat{n}_{\text{spacing}} = \text{median}\{\widehat{n}_i\}.$$

6 Relationship Between Rank Inversion and Spacing Estimators

The rank inversion and spacing estimators, though derived differently, are deeply connected. Both arise from the same underlying relationship between the order statistics of a sample and the cumulative distribution function (CDF) of the population.

6.1 Continuous Framework

Consider a continuous population distribution with cumulative function $F(x)$ and density $f(x) = F'(x)$. Let the order statistics of a sample of size n be

$$X_{(1)} \leq X_{(2)} \leq \cdots \leq X_{(n)}.$$

The expected population rank associated with $X_{(i)}$ is approximately

$$E[\text{Rank}(X_{(i)})] \approx nF(X_{(i)}),$$

so that the rank inversion estimator follows from solving for n:

$$\widehat{n}_i^{(\text{rank})} = \frac{i - 0.5}{F(X_{(i)})}.$$

This estimator depends on the *level* of the cumulative distribution at each observed order statistic.

6.2 Rank Function and Its Derivative

Define the continuous rank function

$$r(x) = nF(x),$$

representing the expected number of population elements less than or equal to x. For the i-th order statistic, $r(X_{(i)}) \approx i$. Differentiating with respect to i gives

$$\frac{dr(X_{(i)})}{di} = nf(X_{(i)})\frac{dX_{(i)}}{di} = 1,$$

which rearranges to

$$\frac{dX_{(i)}}{di} = \frac{1}{nf(X_{(i)})}.$$

Thus, the rate of change in the order statistic with respect to rank index is inversely proportional to both the population size n and the local density $f(X_{(i)})$.

6.3 Spacing Estimator as a Discrete Derivative

In practice, the change in order statistic index is discrete. Replacing the derivative by a finite difference yields

$$\frac{dX_{(i)}}{di} \approx X_{(i+1)} - X_{(i)} = \Delta_i,$$

so that

$$n \approx \frac{1}{\Delta_i f(X_{(i)})}.$$

This is precisely the *spacing estimator*,

$$\widehat{n}_i^{(\text{spacing})} = \frac{1}{\Delta_i f(X_{(i)})}.$$

Hence, the spacing estimator can be viewed as a discrete approximation to the derivative of the rank function.

6.4 Intuitive Explanation

The equation

$$nf(X_{(i)})\frac{dX_{(i)}}{di} = 1$$

captures a simple geometric idea: to move one rank higher in the population, the required change in X depends inversely on the density $f(x)$. When $f(x)$ is large (data are dense), small changes in x correspond to large changes in rank. Conversely, in sparse regions (low $f(x)$), consecutive order statistics are further apart, producing large spacings.

6.5 Discrete Analogy

In a discrete setting, the change in rank between consecutive order statistics is approximately one:

$$\Delta r = n[F(X_{(i+1)}) - F(X_{(i)})] \approx nf(X_{(i)})\Delta_i \approx 1,$$

which rearranges to

$$n \approx \frac{1}{f(X_{(i)})\,\Delta_i},$$

a PDF-based spacing formula, or to

$$n \approx \frac{1}{[F(X_{(i+1)}) - F(X_{(i)})]},$$

which is an equivalent CDF-based spacing formula.

This discrete formulation again leads directly to the spacing estimator, confirming its theoretical link to the rank inversion approach. Moreover, the spacing estimator is just a discrete derivative of the rank inversion estimator. Using a Taylor expansion, for consecutive order statistics $X_{(i)}$ and $X_{(i+1)}$:

$$F(X_{(i+1)}) \approx F(X_{(i)}) + f(X_{(i)})(X_{(i+1)} - X_{(i)}),$$

which can be rearranged to express the population size estimate as

$$n \approx \frac{1}{(X_{(i+1)} - X_{(i)})f(X_{(i)})} \equiv \widehat{n}_i^{(\text{spacing})}.$$

Hence, the rank-inversion estimator effectively accumulates the contributions from all consecutive spacings, since each spacing measures the local increment in population rank:

$$\widehat{n}^{(\text{rank})} \approx \sum_i (X_{(i+1)} - X_{(i)})\, f(X_{(i)}).$$

This highlights that the two estimators are mathematically dual: the spacing estimator captures the local slope of the rank function, while the rank-inversion estimator integrates these local increments to recover the overall population size.

To see more clearly why this holds, we have

$$i - 0.5 = \sum_{j=1}^{i-1} 1 \approx n \sum_{j=1}^{i-1} \big[F(X_{(j+1)}) - F(X_{(j)})\big].$$

Dividing both sides by $F(X_{(i)})$ gives

$$\widehat{n}_i^{(\text{rank})} = \frac{i - 0.5}{F(X_{(i)})} \approx \frac{n \sum_{j=1}^{i-1} [F(X_{(j+1)}) - F(X_{(j)})]}{F(X_{(i)})} \approx \frac{n \sum_{j=1}^{i-1} (X_{(j+1)} - X_{(j)}) f(X_{(j)})}{F(X_{(i)})}.$$

Taking the mean over the central portion of the sample and ignoring the scaling by $F(X_{(i)})$ (which varies slowly in the central region) leads to the approximation

$$\widehat{n}^{(\text{rank})} \approx \sum_i (X_{(i+1)} - X_{(i)})\, f(X_{(i)}),$$

which is exactly the form of the *spacing estimator*.

6.6 Summary of the Relationship

The rank inversion and spacing estimators are two sides of the same underlying functional mapping:

$$r(x) = nF(x), \qquad \frac{dr}{dx} = nf(x), \qquad \frac{dX}{di} = \frac{1}{nf(X_{(i)})}.$$

The rank inversion estimator uses the level of $F(x)$, while the spacing estimator uses its slope $f(x)$. Conceptually, the spacing estimator represents the *derivative* of the rank inversion estimator with respect to order index. This duality provides the mathematical basis for combining them in a hybrid estimator, which balances the stable but biased rank inversion with the locally sensitive but variable spacing method.

7 Robust Aggregation

7.1 Median and MAD

The MATLAB script computes the *median* of multiple repeated sample estimates to reduce sensitivity to outliers:

$$\widehat{n}_{\text{median}} = \text{median}\{\widehat{n}^{(r)}\}_{r=1}^{n_{\text{reps}}}.$$

To quantify variability, the *median absolute deviation (MAD)* is used:

$$\text{MAD} = \text{median}\big|\widehat{n}^{(r)} - \widehat{n}_{\text{median}}\big|.$$

A robust confidence interval is then:

$$\text{CI} = \widehat{n}_{\text{median}} \pm \text{MAD}.$$

7.2 Advantages

- The median is robust against skewed distributions.
- MAD provides a simple, outlier-resistant measure of dispersion.

8 Bias and Variability of the Estimators

The performance of an estimator can be assessed through its *bias* and *variability*. Let $\widehat{n}^{(r)}$ denote the estimate of n obtained from the r-th repetition.

$$\mathrm{Bias}(\widehat{n}) = \mathbb{E}[\widehat{n}] - n_{\mathrm{true}}, \tag{1}$$
$$\mathrm{Var}(\widehat{n}) = \mathbb{E}[(\widehat{n} - \mathbb{E}[\widehat{n}])^2]. \tag{2}$$

In simulations, these quantities are approximated using the replicates:

$$\widehat{\mathrm{Bias}} = \bar{n} - n_{\mathrm{true}}, \quad \widehat{\mathrm{Var}} = \frac{1}{n_{\mathrm{reps}} - 1} \sum_{r=1}^{n_{\mathrm{reps}}} (\widehat{n}^{(r)} - \bar{n})^2, \tag{3}$$

where $\bar{n}$ is the mean of the repeated estimates.

To provide a robust alternative that is less sensitive to outliers, the MATLAB code also computes:

$$\widehat{\mathrm{Bias}}_{\mathrm{robust}} = \mathrm{median}(\widehat{n}^{(r)}) - n_{\mathrm{true}}, \tag{4}$$
$$\widehat{\mathrm{Var}}_{\mathrm{robust}} = \mathrm{MAD}(\widehat{n}^{(r)})^2, \tag{5}$$

where MAD is the median absolute deviation of the repeated estimates.

8.1 Understanding Bias in the Rank Inversion Estimator

The *rank inversion estimator* can exhibit systematic bias, particularly when the sample size m is small relative to the population n. Sources of bias include:

1) **Nonlinearity of CDF inversion:** The estimator assumes $F(X_{(i)}) \approx i/n$. Small deviations in $F(X_{(i)})$ are magnified when inverted, leading to over- or underestimation.

2) **Finite-sample and edge effects:** Extreme order statistics $(X_{(1)}, X_{(m)})$ are highly variable. Even trimming to the central 25%–75% does not fully remove residual bias.

3) **Dependence on assumed distribution:** If the assumed CDF (e.g., Normal or Pareto) differs from the true distribution, the inversion introduces multiplicative bias.

Overall, the rank inversion estimator is often *biased but relatively stable*. Pooling or bootstrap-based debiasing can mitigate, but not eliminate, the bias.

8.2 Understanding Variability in the Spacing Estimator

The *spacing estimator* generally has low average bias but high variance, due to the stochastic nature of inter-point spacings:

1) **Dependence on local density:** The estimator

$$\widehat{n}_i = \frac{1}{\Delta_i f\left(\frac{X_{(i)}+X_{(i+1)}}{2}\right)}, \quad \Delta_i = X_{(i+1)} - X_{(i)},$$

is highly sensitive to small variations in Δ_i and $f(x)$, especially in regions where the PDF is low.

2) **Amplification of noise:** Small estimation errors in Δ_i or $f(x)$ are amplified multiplicatively, producing heavy-tailed sampling distributions for $\widehat{n}_i$.

3) **Intrinsic variability of spacings:** Even with a perfectly known PDF, spacing variance scales roughly as $1/(n^2 f^2(x))$, resulting in large sample-to-sample fluctuations.

Pooling results from multiple repetitions or summarizing using robust statistics (median and MAD) substantially improves stability.

8.3 Bias–Variance Tradeoff

The two estimators illustrate complementary trade-offs:

Estimator	Bias	Variability
Rank Inversion	Moderate–High (systematic)	Low–Moderate
Spacing	Low (on average)	High

- Rank inversion benefits most from bias-reduction strategies such as pooling or bootstrap debiasing.
- Spacing benefits from robust aggregation to control its high variance.
- Pooling across independent samples improves median estimates without violating $m \ll n$.

9 Pooling Samples

Pooling effectively reduces variance without introducing significant bias, as shown in Section 8.

9.1 Concept

Pooling combines estimates from multiple independent small samples before computing the median. This increases the effective sample size without violating the assumption $m \ll n$.

9.2 Implementation

Let R small samples be drawn, each of size m. For each sample, compute $\widehat{n}^{(r)}$ using rank inversion or spacing. Pool all $\widehat{n}^{(r)}$ together and compute the median for a final estimate:

$$\widehat{n}_{\text{pooled}} = \text{median}\{\widehat{n}^{(r)}\}_{r=1}^{R}.$$

9.3 Benefits

- Reduces estimator variance.
- Mitigates the effect of extreme sample realizations.
- Can be used adaptively, stopping when the relative change in the pooled median or the width of the robust confidence interval falls below a threshold.

10 Ratio of Order Statistic Spacings for Different Sample Sizes

Recall that $X_1, \ldots, X_n$ be independent and identically distributed (i.i.d.) random variables from a continuous distribution with strictly positive and differentiable density $f(x) > 0$. Denote the order statistics by

$$X_{(1)} < X_{(2)} < \cdots < X_{(n)}.$$

Consider a smaller sample of size m ($m \ll n$), denoted by

$$Y_1, \ldots, Y_m \sim \text{i.i.d. from the same distribution}, \quad Y_{(1)} < \cdots < Y_{(m)}.$$

Define the spacing between consecutive order statistics in each sample:

$$s_{n,i} = X_{(i+1)} - X_{(i)}, \quad s_{m,i} = Y_{(i+1)} - Y_{(i)}.$$

We are interested in the ratio of typical spacings.

10.1 Theorem (Ratio of Spacings)

For interior order statistics (away from the edges), the expected spacing satisfies

$$\mathbb{E}[s_{m,i}] \approx \frac{n}{m}\,\mathbb{E}[s_{n,j}], \tag{6}$$

where $s_{n,j}$ and $s_{m,i}$ are matched to corresponding quantiles in the central region.

10.2 Proof

1. **Uniform transformation.** Let $U_k = F(X_k)$ and $V_l = F(Y_l)$. Then

$$U_{(1)} < \cdots < U_{(n)}, \quad V_{(1)} < \cdots < V_{(m)}$$

are order statistics from a Uniform$[0,1]$ distribution. Define uniform spacings:

$$\Delta_{n,j} = U_{(j+1)} - U_{(j)}, \quad \Delta_{m,i} = V_{(i+1)} - V_{(i)}.$$

2. **Mean uniform spacings.** For central order statistics, the expected spacing is approximately

$$\mathbb{E}[\Delta_{n,j}] \approx \frac{1}{n}, \quad \mathbb{E}[\Delta_{m,i}] \approx \frac{1}{m}.$$

3. **Mapping back to original scale.** On the original scale, by Taylor expansion (delta method):

$$s_{n,j} \approx \frac{\Delta_{n,j}}{f(X_{(j)})}, \quad s_{m,i} \approx \frac{\Delta_{m,i}}{f(Y_{(i)})}.$$

4. **Take expected values.** Assuming central quantiles so that $f(X_{(j)}) \approx f(Y_{(i)}) \approx f(x)$, we have

$$\mathbb{E}[s_{n,j}] \approx \frac{\mathbb{E}[\Delta_{n,j}]}{f(x)} \approx \frac{1}{nf(x)}, \quad \mathbb{E}[s_{m,i}] \approx \frac{\mathbb{E}[\Delta_{m,i}]}{f(x)} \approx \frac{1}{mf(x)}.$$

5. **Compute the ratio.**

$$\frac{\mathbb{E}[s_{m,i}]}{\mathbb{E}[s_{n,j}]} \approx \frac{1/(mf(x))}{1/(nf(x))} = \frac{n}{m}.$$

Therefore, for interior spacings, the expected spacing of a smaller sample of size m is approximately n/m times the spacing of the larger sample of size n.

□

10.3 Remark

1) This result relies on the smoothness and positivity of f in the central region.

2) Edge spacings may not satisfy this ratio due to boundary effects.

3) The approximation becomes exact in the limit of large m, n with $m \ll n$.

11 Capture-Recapture Ratios

For two consecutive samples of size m, let I denote the number of shared elements. The *capture-recapture ratio* is

$$R = \frac{m+1}{I+1}.$$

In expectation, $R \approx n/m$.

Bias of capture–recapture estimator. It is important to note that the simple estimator based on the overlap,

$$\widehat{n} = \frac{m^2}{I},$$

is nonlinear in the overlap I and therefore biased: since $\mathbb{E}[1/I] \neq 1/\mathbb{E}[I]$, the estimator tends to overestimate n when $m \ll n$ (the overlap can be very small and its reciprocal is heavy-tailed). A standard small-sample correction due to Chapman substantially reduces this bias:

$$\widehat{n}_{\text{Chapman}} = \frac{(m+1)(m+1)}{I+1} - 1,$$

which is nearly unbiased for typical sampling regimes and is widely recommended in practice. When multiple repeated samples are available, one may compute pairwise Chapman estimates and aggregate them (for example by the median); increasing the number of repetitions mainly reduces variance, not the inherent small-sample bias of an individual pairwise estimator. For further details and derivations of the bias and variance of these estimators, see the references below.

Repeated sampling allows estimation of n by averaging R over many repetitions. Combining the median with MAD provides a robust estimator that corrects for extreme fluctuations in overlap.

We menton that the spacing estimator can be multiplied I/m, or $(I+1)/(m+1)$, as a correction, since the ratio is inversely related to the spacing; see Section 10.

References.

1. Chapman, D. G. (1951). Some properties of the hypergeometric distribution with applications to zoological censuses. University of California Publications in Statistics, 1(7), 131–160.

2. Seber, G. A. F. (1982). The Estimation of Animal Abundance and Related Parameters (2nd ed.). Macmillan.

12 Repeated Sampling and Spacing Estimation

Let n denote the population size and $m \ll n$ the size of each sample. In the standard capture-recapture (CR) framework, consider two independent samples of size m, with I elements in common. A classical CR estimate of n is

$$\widehat{n} = \frac{m^2}{I}.$$

To improve robustness and reduce variance, one can draw R independent samples of length m and compute the corresponding $\widehat{n}$ for each pair of samples. From these repeated estimates, one can calculate the average, median, minimum, maximum, and variance, providing a fuller description of uncertainty. Note that in principle there are $n(n-1)/2$ pairs of individuals available for CR estimation, although in practice we work with the sampled elements.

This information can be incorporated into spacing estimation. Recall that the spacing estimator relies on the typical distance between consecutive order statistics. The key observation is that the estimated population-to-sample ratio

$$\frac{\widehat{n}}{m} \approx \frac{m}{I}$$

provides a scaling factor for the spacing. That is, empirical spacings measured in the sample should be multiplied by m/I (or equivalently divided by $\widehat{n}/m$) to produce a more accurate estimate of the population spacing. This simple adjustment accounts for the under-sampling inherent in $m \ll n$ and leverages repeated CR estimates to stabilize the spacing estimate.

12.1 Clarifying the role of repeated sampling

Let n be the population size and $m \ll n$ the sample size. For two independent samples of size m, let I denote their intersection. The capture-recapture estimate $\widehat{n} = m^2/I$ approximates the population-to-sample ratio n/m. Repeatedly drawing R independent samples allows computation of multiple $\widehat{n}$ values. Aggregating these via the mean or median reduces variance and stabilizes the estimate, providing a robust scaling factor for spacing estimation. Specifically, the central sample spacings should be divided by the estimated ratio $m/I \approx \widehat{n}/m$ to correct for finite-sample effects. Across repeated samples, the distribution of $\widehat{n}$ (e.g., variance, interquartile range, or MAD) quantifies the uncertainty and guides robust estimation. Although only a small fraction of the $n(n-1)/2$ possible pairs is observed, repeated sampling effectively approximates the population-level spacing statistics. This demonstrates how repeated sampling improves both the accuracy and robustness of spacing-based estimators.

13 Simulation and Summary Tables

To illustrate the estimators' behaviour, synthetic populations of size $n_{\text{true}} = 10{,}000$ were generated under both normal and Pareto distributions. For each population, $m = 1000$ samples were drawn repeatedly ($n_{\text{reps}} = 500$), and both the rank inversion and spacing estimators were computed for each replicate.

For each repetition:

1. Draw a sample of size m.
2. Compute $\widehat{n}_{\text{rank}}$ and $\widehat{n}_{\text{spacing}}$ using the methods described in Sections 4 and 5.
3. Aggregate results over repetitions using robust statistics: median and MAD.

These simulations allow direct evaluation of the *bias* and *variability* of the estimators, as discussed in Section 8. In particular, the tables below summarize:

- Mean and median of the repeated estimates.
- Standard deviation and MAD.
- Absolute bias relative to the true population size n_{true}.
- Robust confidence intervals (median $\pm$ MAD).

13.1 Example Results from MATLAB Simulation

Tables 2 and 3 show results for normal and Pareto populations, respectively, using the simulation parameters in Table 1. These results illustrate the complementary bias–variance characteristics of the rank inversion and spacing estimators, and demonstrate the stabilizing effect of robust aggregation.

Table 1: MATLAB simulation Parameters

Parameter	Value
True population size, n_{true}	10000
Normal mean, μ_{true}	0
Normal standard deviation, σ_{true}	2
Pareto shape, α	2.5
Pareto scale, x_m	1
Sample size, m	1000
Number of repetitions, n_{reps}	500

Table 2: Rank and Spacing Estimator Results for Normal Population

Estimator	Mean	Median	Std	MAD	Med+MAD	AbsBias	CI Lower	CI Upper
Rank	999.70	999.37	11.73	7.45	1006.82	8993.18	991.92	1006.82
Spacing	10950.69	7509.16	16074.34	2032.95	9542.11	457.89	5476.20	9542.11

Table 3: Rank and Spacing Estimator Results for Pareto Population

Estimator	Mean	Median	Std	MAD	Med+MAD	AbsBias	CI Lower	CI Upper
Rank	1001.07	1001.00	28.31	19.31	1020.31	8979.69	981.69	1020.31
Spacing	15621.84	7476.17	73121.07	2156.81	9632.97	367.03	5319.36	9632.97

14 Discussion

- Both methods rely on order statistics to infer the fraction of the population represented by the sample.
- Rank inversion uses the parametric CDF to relate order statistics to population fraction.
- Spacing uses the PDF and the distance between consecutive order statistics.
- Using only central 25%–75% of the sample mitigates edge effects.
- Pooling multiple small samples improves robustness and reduces variance.

- The MAD provides a simple, robust measure of estimator uncertainty.
- The median+MAD estimate corrects for extreme variations in sample overlap, providing more reliable estimates for the higher-variance spacing estimator.

15 Hybrid Estimation with Penalised Weighting

To combine the inverse-rank and spacing estimators for the population size, we define a hybrid estimator as

$$\widehat{n}_H = w\widehat{n}_1 + (1-w)\widehat{n}_2,$$

where $\widehat{n}_1$ is the inverse-rank estimator, $\widehat{n}_2$ is the spacing estimator, and $w \in [0,1]$ is a weighting coefficient. The objective is to minimise a penalised mean squared error (MSE) criterion that balances bias, variance, and regularisation:

$$\mathrm{MSE}_H(w) = \mathrm{Var}(\widehat{n}_H) + [\mathrm{Bias}(\widehat{n}_H)]^2 + \lambda w^2,$$

where $\lambda \geq 0$ is a penalty parameter controlling the relative influence of the inverse-rank estimator.

15.1 Derivation of the Optimal Weight

Let $b_1 = \mathbb{E}[\widehat{n}_1] - n$, $b_2 = \mathbb{E}[\widehat{n}_2] - n$ denote the biases, and $v_1 = \mathrm{Var}(\widehat{n}_1)$, $v_2 = \mathrm{Var}(\widehat{n}_2)$ the variances of the two estimators. Denote by ρ their correlation. The expectation and variance of the hybrid estimator are

$$\mathbb{E}[\widehat{n}_H] = w\mathbb{E}[\widehat{n}_1] + (1-w)\mathbb{E}[\widehat{n}_2],$$

$$\mathrm{Var}(\widehat{n}_H) = w^2 v_1 + (1-w)^2 v_2 + 2w(1-w)\rho\sqrt{v_1 v_2}.$$

Substituting these into the penalised MSE and minimising with respect to w gives

$$\boxed{w^* = \frac{v_2 + b_2(b_2 - b_1) - \rho\sqrt{v_1 v_2}}{v_1 + v_2 + (b_1 - b_2)^2 - 2\rho\sqrt{v_1 v_2} + 2\lambda}.}$$

This expression defines the optimal weight w^* under the penalised MSE criterion. The final hybrid estimator is then

$$\widehat{n}_H = w^*\widehat{n}_1 + (1-w^*)\widehat{n}_2.$$

15.2 Interpretation of the Penalty Parameter λ

The penalty parameter λ acts as a *regularisation term* that controls the stability of the hybrid estimator. When λ is small, the estimator fully adapts to the empirical bias and variance estimates, which may be sensitive to sampling noise. As λ increases, the weight w^* is progressively shrunk towards zero, reducing the influence of the inverse-rank estimator and leaning more heavily on the spacing estimator:

$$\lambda \to 0 \quad \Rightarrow \quad w^* \approx w_{\mathrm{MSE}},$$

$$\lambda \to \infty \quad \Rightarrow \quad w^* \to 0.$$

Thus, increasing λ systematically shifts the balance towards the spacing estimator, which typically has higher variance but lower bias.

15.3 Optimising the Penalty Parameter

The penalty λ can be tuned using one of the following approaches:

1. **Cross-validation:** Partition the simulation replicates into training and validation subsets. Choose λ that minimises the validation MSE of the hybrid estimator:

$$\lambda^* = \arg\min_{\lambda} \mathrm{MSE}_{\mathrm{val}}(\widehat{n}_H(\lambda)).$$

2. **Analytic heuristic:** A simple practical rule is

$$\lambda^* \approx c\frac{v_1 - v_2}{n_{\mathrm{true}}^2}, \qquad c \in [0.1, 2],$$

   which penalises high-variance estimators more strongly.

3. **Empirical tuning:** In simulation studies, evaluate MSE for a grid of λ values (e.g. $\lambda \in \{0.1, 0.5, 1, 2, 5\}$) and select the value corresponding to the "elbow" of the MSE curve.

15.4 Theoretical Interpretation and Connections

The hybrid estimator can be viewed as a form of *shrinkage estimation*, in the spirit of ridge regression or the James–Stein estimator. The penalty term λw^2 introduces a bias–variance trade-off that discourages extreme weights when the estimators are noisy or weakly correlated. In this view:

- The inverse-rank estimator acts as a low-variance, high-bias estimate.
- The spacing estimator serves as a stable "anchor" with higher variance but lower bias.

The optimal weight w^* therefore represents a data-driven *shrinkage factor* that pulls the combined estimator towards the more stable spacing-based estimate when uncertainty is high. Increasing λ strengthens this shrinkage, improving robustness at the cost of potential underestimation. This behaviour is directly analogous to how ridge regression penalises large coefficients or how the James–Stein estimator shrinks noisy estimates towards a grand mean.

15.5 Summary

The hybrid estimator provides a principled compromise between two competing estimators:

- the inverse-rank estimator — typically high bias but low variance;
- the spacing estimator — typically lower bias but higher variance.

The penalty parameter λ directly controls the trade-off between these two sources of error, allowing for adaptive balance and improved stability across repeated sampling experiments.

16 Conclusion

Order-statistics-based estimators provide an effective way to estimate the size of a large population from small samples, especially when the underlying distribution is known or well approximated. Robust aggregation techniques like median and MAD, along with pooling, allow practical application in real-world scenarios where repeated sampling is feasible. The spacing estimator trades higher variance for lower bias, while combining median and MAD yields a robust estimate suitable for finite samples.